

LiMNet: Lightweight Temporal Unimodal Radar for 3D Object Detection

Maarten Strebe

Hunor Kovács

Abstract

Robust 3D object detection typically relies on LiDAR and cameras, which degrade in adverse weather. While automotive radar is weather-resilient, its inherent sparsity often restricts it to multi-modal fusion pipelines, severely limiting the use of geometric data augmentations during training. To unlock radar’s standalone potential, we introduce LiMNet, a lightweight unimodal detection pipeline. By “unshackling” the network from fusion constraints, LiMNet leverages temporal multi-frame accumulation and spatial augmentations to achieve a competitive 72.06[%] mAP (ROI) on the View of Delft dataset. Additionally, we investigate whether the lack of dense radar geometry requires explicit mitigation through three manually engineered features: local point density, temporal gradients, and shadow channeling. Our ablation studies demonstrate that these handcrafted features actually worsen performance and add unnecessary computational load. We conclude that standard voxel encoders are already capable of extracting complex spatio-temporal dynamics directly from the data, which makes manual feature engineering redundant for unimodal radar perception.

1. Introduction

Robust 3D object detection is crucial for autonomous driving. While LiDAR and cameras achieve state-of-the-art performance, their reliability degrades heavily in adverse weather conditions [2]. Automotive radar provides a weather-resilient alternative. However, its sparsity and inherent noise typically forces the sensor into a supporting role in multi-modal fusion architectures. Secondly, this reliance on sensor fusion introduces strict spatial constraints, which prohibit the use of aggressive geometric augmentations during training [6]. As a result, automotive radar in this context remains underdeveloped compared to its potential.

To overcome these limitations, we propose a lightweight, unimodal radar detection pipeline. By isolating the radar and “unshackling” the network from camera-fusion constraints, we are able to use aggressive spatial augmenta-

tions alongside multi-frame accumulation. We demonstrate that this approach allows a standard CenterPoint architecture [10] to achieve competitive performance on the View-of-Delft dataset [7], proving that standard voxel encoders can effectively process sparse temporal radar.

In addition to our baseline pipeline, we investigate the impact of manual feature engineering on sparse radar representations. To compensate for the lack of dense structural geometry, we implement and evaluate three explicitly engineered spatial features designed to capture velocity, temporal acceleration, and multi-path occlusion.

2. Related work

We start by reviewing recent advancements in LiDAR-based 3D object detection and radar sensor fusion, and then focus on temporal point cloud processing techniques to contextualize our proposed unimodal radar approach.

2.1. LiDAR-Based 3D Object Detection.

LiDAR point clouds have driven rapid advancements in 3D object detection through voxel-based architectures like VoxelNet [12], efficient column-based encoders like PointPillars [3], and center-based representations like CenterPoint [10]. While achieving state-of-the-art performance, their reliance on dense spatial geometry causes them to struggle when applied directly to highly sparse radar sensors or during adverse weather conditions [4].

2.2. Radar Perception and Sensor Fusion.

Radar provides a weather-resilient alternative to optical sensors. While multimodal datasets [1, 7] have accelerated radar research, most literature still treats it merely as an secondary sensor within fusion pipelines (e.g., RadarNet [9]). This heavy reliance on fusion introduces a critical bottleneck: the strict spatial calibration required between the sensors severely limits the application of geometric data augmentations [11]. Consequently, unimodal radar detection remains underdeveloped due to the sensor’s inherent sparsity and noise.

2.3. Temporal Point Cloud Processing.

Because single radar sweeps provide limited context, temporal accumulation is standard practice. Traditional architectures [3, 10] achieve this simply by concatenating sweeps and appending relative timestamps as scalars. Conversely, advanced approaches deploy computationally heavy modules, such as memory-based RNNs [8] or temporal relational layers [5]. These approaches introduce heavy computational overhead or treat time as a simple scalar, neglecting the direct spatial changes caused by physical object displacement.

2.4. Our Contribution.

Unlike previous works relying on complex temporal networks [5] or multi-sensor calibration [9], we propose a lightweight unimodal radar solution.

1. By isolating the radar and removing the strict spatial calibration constraints inherent in multi-sensor fusion, we enable the use of aggressive geometric data augmentations on temporal accumulated frames.
2. We establish a competitive detection baseline by utilizing multi-frame accumulation and a standard voxel encoder, avoiding the need for complex temporal networks or multi-sensor calibration.
3. We conduct an ablation study to determine the necessity of manual feature engineering in sparse point clouds (local density, temporal gradients, and shadow channeling).

3. Methodology

In this section we discuss our method for improvement of object detections using radar point cloud data, adapting the CenterPoint architecture introduced by Yin (2021) [10], originally developed for LiDAR point clouds. Our approach consists of improvements on the baseline from data pre-processing to the optimization architecture.

3.1. Frame Transformation.

Because the usage of CenterPoint is on the LiDAR domain, our first improvement consists of transforming the radar point cloud data from the coordinate frame of its sensor to the LiDAR frame of reference. This step is essential, since our ground truth bounding boxes are in the coordinate system of the LiDAR sensor. This ensures that the radar points representing each object align accurately with its 3D bounding box, thereby training the model with accurate loss values.

3.2. Temporal Accumulation.

One of the reasons radar data is challenging, especially in a framework optimized for LiDAR point clouds, is the sparsity of points that the radar provides. One of our methods of addressing this difficulty is in the accumulation of temporal radar data. The past five point cloud frames are fused into one with compensation applied for the ego-motion of the vehicle. This makes for increased point density of the point cloud based on the relative velocity of objects to the sensor.

3.3. Data Augmentation.

Since our detector relies solely on unimodal radar data, we have fewer augmentation constraints than a camera-radar early-fusion model, allowing us to apply more aggressive augmentation methods. The point cloud augmentation consists of two parts. First, we apply a random rotation with a zero-mean distribution around the Z-axis. Then with a certain probability, we also apply a mirroring operation on the XZ-plane of the radar frame.

3.4. Optimizer Improvements.

To address stability and overfitting, we utilize the AdamW optimizer with decoupled weight decay for improved regularization. Together with a learning rate scheduler, enabling the network to efficiently traverse the loss landscape.

3.5. Manual Feature Engineering.

To evaluate whether sparse radar requires explicit manual feature engineering, we engineer three features prior to voxelization. These are formulated as hypotheses for our ablation studies:

- **Local Point Density:** We hypothesize that higher local density within our point cloud helps distinguish a valid object detection from the environment. We calculate the average Euclidean distance to the $k = 5$ nearest neighbors and append this scalar to provide an explicit geometric representation of density.
- **Temporal Density Gradient:** The point cloud density metric, integrated with the temporal frame accumulation can capture the velocity of objects. We hypothesize this gradient explicitly encodes radial and tangential motion vectors to help isolate dynamic targets.
- **Shadow Channeling:** radar waves cannot penetrate dense metal, meaning a second return directly behind high-RCS points are likely to be ghost detections. We formulate a binary "shadow" feature for points in these occluded regions. We hypothesize this will explicitly teach the network to suppress this form of noise.

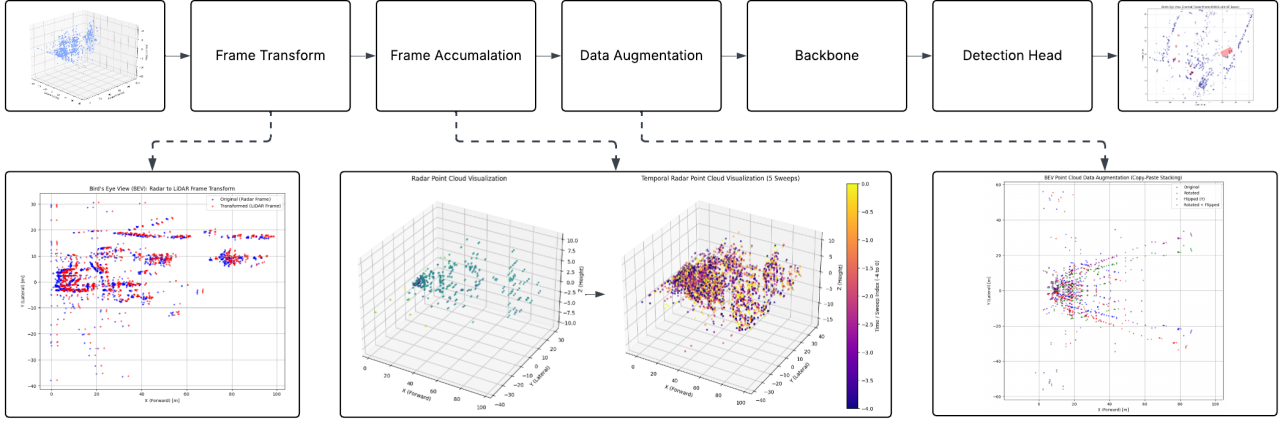


Figure 1. Overview of the LiMNet architecture.

4. Experiments

In this section, we evaluate our unimodal radar detection pipeline, LiMNet. We first detail the experimental setup and implementation details. We then present the overall performance of LiMNet compared to the baseline CenterPoint structure, followed by ablation studies that isolate the impact of our pipeline modifications and manually engineered features.

4.1. Experimental Setup

Dataset and Metrics. We evaluate our approach using the View of Delft (VoD) dataset, which provides synchronized 3D radar, LiDAR, and camera data in urban environments. Following standard evaluation protocols, our primary metric is the mean Average Precision (mAP), reported both for the Region of Interest (ROI) and the entire recorded area. We evaluate all hyperparameter tuning and ablation studies, including all data reported in Table 1 and Table 2, exclusively on the VoD validation set. All models are trained for 12 epochs.

Implementation Details. To adapt the LiDAR-centric CenterPoint architecture for sparse radar, we tune several key hyperparameters. The optimal configuration for LiMNet, established through empirical searches is summarized in Table 1. All further ablation studies are conducted using these optimally tuned parameters to isolate each component’s contribution.

4.2. Results

The standard CenterPoint architecture applied to single-sweep radar data achieves an overall mAP of 47.74% within the ROI. By implementing our optimized pipeline, LiMNet achieves a final performance of 66.47% mAP (ROI) and 45.67% (Entire Area). This absolute improvement of

Hyperparameter	Optimal Setting
Optimization	
Voxel Resolution	0.16 m
Optimizer	AdamW
Learning Rate	0.001
Gradient Clipping	2.0
Augmentation	
Z-Axis Rotation	$[-20^\circ, +20^\circ]$
XZ-Plane Mirroring	50%

Table 1. Summary of optimal implementation details and data augmentation strategies used to construct the LiMNet baseline.

18.73% mAP over the baseline demonstrates the efficacy of our approach.

The contributions of each feature are summarized in Table 2. With the alignment of bounding boxes and points the mAP of detections in the ROI increased by 7.4%. Temporal multi-frame accumulation further contributed to the results by nearly 8%. Furthermore improving the optimizer amounted to another 0.5%. The initial rotational augmentation improves the performance on the region of interest by 0.87%, while the subsequent mirror augmentation contributes to the improvement just under 2%. We further investigate the augmentation by setting the ratio of mirrored samples to 100%. This results in an mAP decrease in the region of interest of 3.3% compared to the baseline 50% ratio.

After the initial increase of 19 minutes, frame accumulation adds a further 9 minutes to the training time. The optimizer, by contrast, reduces training time by 11 minutes. Rotational augmentation contributes an additional 27 minutes, and the final mirroring augmentation adds 39 minutes,

Method	mAP (ROI) [%]	mAP (Entire) [%]	Final Loss	Time [h]
Baseline	[47.73]	[26.00]	[7.11]	[0:51]
+ Frame Transform	[55.12]	[34.07]	[18.48]	[1:10]
+ Accumulation	[63.11]	[42.08]	[5.74]	[1:19]
+ Optimizer Improvements	[63.62]	[42.68]	[5.59]	[1:08]
+ Rotation Augmentation	[64.49]	[44.30]	[7.42]	[1:35]
+ Mirroring Augmentation	[66.47]	[45.67]	[6.67]	[2:14]
<i>Evaluation of Manual Features</i>				
+ Local Point Density	[65.89]	[43.20]	[6.81]	[2:42]
+ Shadow Channeling	[65.78]	[42.98]	[6.83]	[2:59]
+ Temporal Density Gradient	[66.39]	[43.68]	[6.65]	[2:13]

Table 2. Incremental ablation study on the View of Delft validation set. The top section demonstrates the construction of our competitive augmented baseline. The bottom section details the negative results of explicitly engineering physical features.

bringing the total training time to 2 hours and 14 minutes.

As shown in the bottom section of Table 2, incorporating the handcrafted spatial features proposed in Section 3 yields negative results across all metrics. The local point density metric results in a degradation to 65.89% mAP while heavily increasing the training time from 2:14 to 2:42 hours. Similarly, the shadow channeling feature drops the mAP to 65.78% and increases the training time to nearly three hours. Similarly the Temporal Density Gradient decreases performance and training time by negligible amounts.

To validate generalization, we evaluate the optimized LiMNet model on the unseen test set, achieving a highly competitive ROI mAP of 72.06%.

4.3. Analysis and Discussion

The improvement over the baseline validates our unimodal strategy. The frame accumulation and homogeneous transformation contribute most to detection performance in absolute terms; however, as these are implemented first, subsequent techniques may have less room for improvement.

While the augmentation triples the dataset size, it causes training time to grow by less than twofold, reflecting an overall improvement in training efficiency. The optimizer improvement, while increasing the mAP to a lesser extent, amounts to the reduction of training time, counterbalancing increases by other methods.

The results of the Manual Features indicate that standard voxel encoders already learn to approximate complex spatial dynamics, making explicit feature engineering redundant.

Despite a highly competitive test set performance, purely unimodal radar faces fundamental challenges. The model excels on high-RCS (Radar Cross Section) targets like cars, but struggles with pedestrians due to the extreme point sparsity of low-reflective human bodies. Additionally, our qual-

itative analysis shows the network is occasionally deceived by reflections from large metallic surfaces, creating "ghost" point clusters.

5. Conclusion

In this paper we present LiMNet, a highly competitive unimodal radar 3D object detector. By intentionally isolating the radar modality from the strict constraints of multi-sensor fusion, we enable the use of spatial augmentations on a five-frame temporal accumulation pipeline. This unconstrained approach successfully improves upon the baseline by almost 19% absolute mAP on the validation set and a final performance of 72.06% mAP on the View of Delft test set. Additionally, we evaluate explicitly engineered spatial features. Our empirical results demonstrate that these features degrade both detection accuracy and computational efficiency. We conclude that when supported by geometric augmentation and temporal context, standard voxel encoders implicitly learn complex spatial dynamics, making manual feature engineering unnecessary for sparse radar perception. Future work focuses on addressing the inherent physical limitations of unimodal radar to suppress "ghost" reflections and improve the detection of low-RCS road users such as pedestrians.

6. Contributions

Both authors contributed equally to the overall conceptualization, design, and final revision of the assignment. **Maarten Strebe** implemented the optimizer and the manual feature engineering. Additionally wrote the Abstract, Introduction, and Conclusion, and contributed to the Methodology and Experiments sections. **Hunor Kovács** developed the temporal accumulation pipeline and the augmentations, and contributed to the Methodology and Experiments sections.

References

- [1] Holger Caesar, Varun Bankiti, Alex H. Lang, Sourabh Vora, Venice Erin Liong, Qiang Xu, Anush Krishnan, Yu Pan, Giancarlo Baldan, and Oscar Beijbom. nuscenes: A multi-modal dataset for autonomous driving. In *CVPR*, 2020. 1
- [2] Jules Karangwa, Jun Liu, and Zixuan Zeng. Vehicle detection for autonomous driving: A review of algorithms and datasets. *IEEE Transactions on Intelligent Transportation Systems*, 24(11):11568–11594, 2023. 1
- [3] Alex H. Lang, Sourabh Vora, Holger Caesar, Lubing Zhou, Jiong Yang, and Oscar Beijbom. Pointpillars: Fast encoders for object detection from point clouds, 2019. 1, 2
- [4] Seungjun Lee. Deep learning on radar centric 3d object detection, 2020. 1
- [5] Peizhao Li, Pu Wang, Karl Berntorp, and Hongfu Liu. Exploiting temporal relations on radar perception for autonomous driving. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022. 2
- [6] Jianan Liu, Qiuchi Zhao, Weiyi Xiong, Tao Huang, Qing-Long Han, and Bing Zhu. Smurf: Spatial multi-representation fusion for 3d object detection with 4d imaging radar, 2023. 1
- [7] Andras Palffy, Ewoud Pool, Srimannarayana Baratam, Julian F. P. Kooij, and Darius M. Gavrilă. Multi-class road user detection with 3+1d radar in the view-of-delft dataset. *IEEE Robotics and Automation Letters*, 7(2):4961–4968, 2022. 1
- [8] Loveneet Saini, Mirko Meuter, Hasan Tercan, and Tobias Meisen. Attentivegru: Recurrent spatio-temporal modeling for advanced radar-based bev object detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pages 2440–2449, June 2025. 2
- [9] Bin Yang, Runsheng Guo, Ming Liang, Sergio Casas, and Raquel Urtasun. Radarnet: Exploiting radar for robust perception of dynamic objects, 2020. 1, 2
- [10] Tianwei Yin, Xingyi Zhou, and Philipp Krähenbühl. Center-based 3d object detection and tracking, 2021. 1, 2
- [11] Yi Zhou, Li Liu, Haipeng back Zhao, Miguel López-Benítez, Limin Yu, and Yong Yue. Towards deep radar perception for autonomous driving: Datasets, methods, and challenges. *Sensors*, 22(11):4208, 2022. 1
- [12] Yin Zhou and Oncel Tuzel. Voxelnet: End-to-end learning for point cloud based 3d object detection. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2018. 1